Understand the City, Improve Life

A Privacy-First Smart City Data Platform

Turning camera footage → safe, useful city data

AI Pipeline · Go Backend · Next.js Map

The Problem

Cities produce millions of images daily — but using this data faces two big obstacles:

- **Privacy risk** — footage contains faces, license plates, personal data (GDPR/KVKK)
- **Raw data is useless** — a video won't tell you "there's a damaged sign here"

> **Question:** How do we understand the city through data *without* violating citizens' privacy?

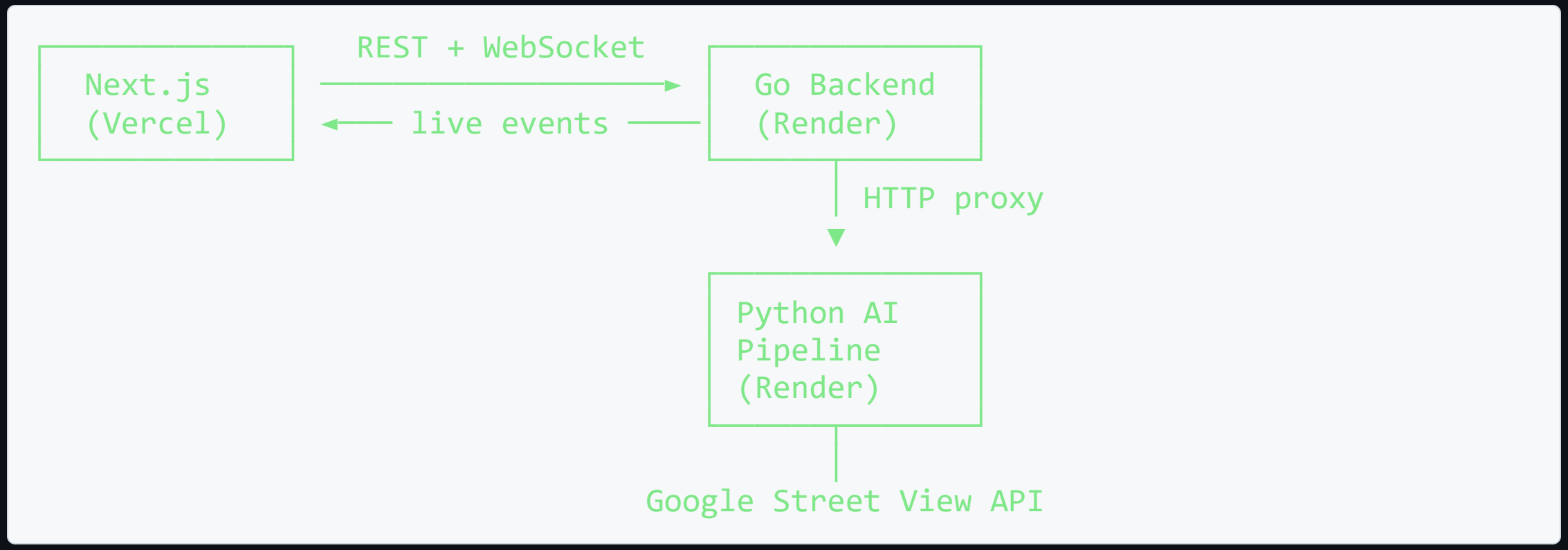
Our Solution

A three-layer, end-to-end **privacy-first** data platform:

```
Footage  →  Anonymize  →  Detect  →  Serve  →  Show on Map
            (face/plate)   (signs)    (API)
```

1. **AI Pipeline** — anonymizes footage, detects urban objects
2. **Go Backend** — filters, enriches, serves data in real time
3. **Next.js Frontend** — displays it on an interactive map

System Architecture



1. AI & Privacy Pipeline

Fail-closed design — if anonymization fails, the pipeline halts and never leaks raw data.

```
Street View footage
  → face + license plate blur   (irreversible)
  → detection ONLY on anonymized frames
  → remove duplicate detections
  → JSON output + raw-data deletion proof
```

- Anonymization with **OpenCV** · detection with **YOLOv8**
- The model **never** sees the original footage

GDPR / KVKK & Privacy (Critical)

> "The model runs on **anonymized** video first, never on raw footage."

Principle	Implementation
Purpose limitation	Urban object detection only — NO identity detection
Mandatory anonymization	Faces + plates irreversibly blurred before training
Data minimization	JSON contains only: type, coordinates, confidence, time
Deletion proof	Raw data auto-deleted and documented

JSON Output — The Only Data We Produce

```json
{
  "type": "damaged_sign",
  "latitude": 41.021,
  "longitude": 28.874,
  "confidence": 0.91,
  "timestamp": "00:01:24"
}
```

No identity. No imagery. Just city data.

2. Go Backend — Production Grade

The **smart bridge** between the AI pipeline and the map:

- **Filterable API** — type, confidence score, geographic bounding box (bbox)
- **Stats endpoint** — type distribution, average confidence
- **WebSocket** — live scan tracking (started → running → completed)
- **Resilience** — rate limiting, panic recovery, graceful shutdown
- **30s cache** — serves data even if the pipeline briefly goes down

Only **3 external packages** — the rest is Go's standard library.

API Endpoints

Method	Path	Description
`GET`	`/health`	Service + pipeline status
`GET`	`/api/v1/detections`	Filtered detection list
`GET`	`/api/v1/detections/stats`	Statistics
`POST`	`/api/v1/scan`	Start a new scan
`GET`	`/api/v1/scan/status`	Scan status
`GET`	`/ws`	WebSocket — live events

Real-Time Experience

User picks a coordinate → scan starts → **watches live**:

```
{ "event": "scan_started",   "payload": {...} }
{ "event": "scan_progress",  "payload": {"status": "running"} }
{ "event": "scan_completed", "payload": {"detection_count": 5} }
```

Thanks to WebSocket, no "loading..." wait — **instant feedback**.

Tech Stack

Layer	Technology
Frontend	Next.js · Vercel
Backend	Go (chi, gorilla/websocket) · Render
AI Pipeline	Python · OpenCV · YOLOv8 · Flask · Render
External API	Google Street View API
Deployment	Docker (multi-stage, ~7 MB image)
Dev Environment	Cursor IDE · agentic ruleset · Git feature branches

Deployment — Live

Both services run live on **Render.com** via Docker:

- **Go Backend** → `cursor-j7jx.onrender.com`
  - `scratch` image · ~7 MB · near-zero attack surface
- **Python Pipeline** → `ai-privacy-pipeline.onrender.com`
  - Gunicorn · health checks · automatic sample fallback

The health endpoint reports both services' status live.

Live Demo

```
# 1. Is the service up?
curl .../health
→ { "status": "ok", "pipeline_status": "ok" }

# 2. Filter damaged signs with high confidence
curl ".../api/v1/detections?type=damaged_sign&min_confidence=0.8"

# 3. Dashboard statistics
curl .../api/v1/detections/stats
→ { "total": 5, "by_type": {...}, "avg_confidence": 0.8 }
```

Then: live scan over WebSocket → pins appearing on the map.

Why This Project Matters

- **Social benefit** — safer cities via damaged-sign and pothole detection
- **Privacy-first** — GDPR/KVKK compliance built into the architecture, not bolted on
- **Data-driven decisions** — real coordinate-based data, not guesswork
- **Scalable** — can grow into a real smart-city service

> Real value begins when code leaves the screen and touches a human life.

Thank You

We welcome your questions

Understand the City · Improve Life · Protect Privacy